

Addressing Lecture Concepts in PetCuesGame

Conceptualization and Work

1. Community of Practice (CoP) and Digital Habitats

We identified Regina Humane Society and schools as early, core Communities of Practice (CoPs). They share a common concern for promoting animal welfare and child protection. The game was designed to serve these virtual habitats by providing an accessible, interactive tool that enhances their education and outreach efforts.

Lecture readings on Digital Habitats (Wenger, White & Smith) had us explore how the communities are using technology to coalesce and share knowledge. This was useful in focusing on lightweight web-based deployment to provide existing digital behavior like online learning portals and cell-phone-based education tools.

2. SDG Alignment and Social Impact

Amongst the major discussions we had during class was how tech solutions approach sustainable development. We addressed this by linking project goals to SDG 3 (Good Health & Well-being), SDG 4 (Quality Education), and SDG 15 (Life on Land). This relevance touched both subject matter (instructing safety, empathy, and care about animals) as well as deployment (making availability on all hardware a priority).

3. Requirements Gathering and Stakeholder Engagement

Using our lecture techniques, such as empathy mapping, persona building, and stakeholder feedback loops, we conducted structured consultations with our client, Rebecca from the Regina Humane Society. Her feedback inspired us to add more animations, streamline character interactions, and include more pet types in order to make it more realistic and interactive.

4. Prototyping and Iteration

We followed the Design Thinking process, as emphasized in lectures:

- Empathize: Spent time with children, teachers, and humane society staff.
- Define: Defined a clear problem: children lack structured, interactive pet safety training.
- Ideate: Created pet interaction situation brainstorming.

- Prototype: Developed a minimum viable product (MVP) with Godot, based on FSM pet behaviors and decision-based game design.
- Test: Continuous testing and feedback loops were employed during development, especially after classroom demos and client feedback.

5. Technology Configuration Inventory

Lecture familiarity with community technology inventories helped us determine what hardware was commonly used in schools and humane society initiatives. This led us to use WebGL cross-device, so the game could be played on tablets, phones, and PCs without the need for specialized hardware.

6. Digital Literacy and 21st Century Skills

Lectures focused on the need for digital learning spaces that support educational content as well as skill acquisition. PetCuesGame promotes:

- Digital literacy through interactive use.
- Critical thinking through decision-based play.
- Empathy and social responsibility through behavioral modeling.

7. Assessment and Feedback Mechanisms

Prompted by discussions of formative assessment tools, we embedded quizzes and immediate feedback into the game. This allowed students to reflect on their choices and replay situations for improved understanding—a basic principle learned in class.

Briefly, the ideas of our course have directly impacted the project from start to finish. From conceptualization and stakeholder engagement to platform choice and educational impact, PetCuesGame is an applied application of our lecture themes, bridging community need with innovative design and educational technology.